Data-Driven Identification of Vessel-Level Fishing Grounds

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Abstract

Knowing the precise location of fishing operations is crucial to managing fish stocks, guiding marine spatial planning and conservation policies, and understanding fisheries' exposure to environmental hazards. While the increased availability of open source vessel-tracking data allow us to track vessels, we still lack a standardized, data-driven method to identify fishing grounds. Here, we develop a heuristic that relies on a single discoverable parameter—the expected daily amount of time a vessel spends in a fishing ground— to identify vessel-level fishing grounds. Our approach relies on two well-known data mining algorithms, the Density-Based Spatial Clustering for Applications with Noise (DBSCAN) and K-Nearest Neighbors, to identify fishing grounds based on vessel-tracking data. Using Mexico's vessel-monitoring systems data, we identify X fishing ground belonging to N fishing vessels that use five different gears. We then perform to applied exercises. First, we quantify habitat utilization and revisitation rates, fidning that X, Y, and Z. We then show how the identified fishing grounds can be used to measure vessel-level exposure to environmental hazards by using hurricane sin the Gulf of Mexico as an example.

1 Introduction

Why do we need to know where vessels fish? from management, knowing where effort is applied tells us where catch comes from. For spatial management, clarity on potential overlap. For fisheries climate-econometrics, gridded data come short. For MPA design, gridded data come short ("equitable designed" / 1% of fleet might be 100% of me). [1, 2]

Previous work on identifying fishing grounds from tracking data: Kroodsma et al. [3] first global footprint (aggregated across vessels); [4] maybe does it on a grid? [5] used Generalized Linear/Additive/Boosting Models (GLM, GAM, GBM, respectively), Classification Tree Analysis (CTA), Artificial Neural Network (ANN), Surface Range Envelope (SRE), Flexible Discriminant Analysis (FDA), Multiple Adaptive Regression Splines (MARS), and RF to build species distribution models (not the same as actual grounds).

Technological advances in vessel-tracking technologies, publicly available data, and improvements in high performance computing now allow us to precisely track the location of thousands of fishing vessels in near-real time. And yet, these data are often summarized and aggregated across spatial and temporal scales imposed by the researcher, loosing important variation in how vessels utilize marine space.

Previous work set defensible but arbitrarily chosen thresholds on the spatial resolution for the identification of fishing grounds. For example, X did A and Y did B. The same is true for temporal thresholds, where a specific number of hours are required for the activity within a grid cell to count. My method differs in a few ways. First, the spatial scale is determined by the data themselves and are a product of the time required for a fishing ground to count as such. Second, because the size of a fishing ground depends on utilization intensity rather than fixed by the researcher, it is possible to perform robustness checks to determine how the chosen utilization density affects it. And, finally, because the utilization density can be easily identified with a simple question posed to fishers: "*When you go out to fish, how much time do you spend in a given place before relocating or coming back to port?*"

2 Data and Methods

2.1 Vessel-Monitoring Systems Data

Vessel-tracking data come from Mexico's Satellite Fisheries Monitoring Program ("Sistema de Monitoreo de Embarcaciones Pesqueras"; sis [6]). The data are pre-procesed through a data standardization, cleaning, and fusion pipeline to make them analysis-ready [7]. The pipeline includes removing observations on land, segmenting tracks, and calculating time spent at each location (defined as the time elapsed since the last ping). The resulting data

set contains the vessel identifier, timestamp, coordinates, and vessel characteristics (fishing gear, target species).

2.2 Data Processing and Sample Construction

We begin by splitting our sample into five main fleets because vessels using different gears and targeting different species move differently and therefore have different density utilizations. For example, a shrimp trawler often operates in smaller, localized areas known to have shrimp, while a tuna purse seiner that needs to scour the seas to find tuna schools. The vessel registry reports the fishing gear and target species for which each vessel holds a license. We use this information to group vessels into five fleets: (1) vessels with bottom trawls and shrimp permit were classified as shrimp trawlers (N = X), (2) vessels with longlines and a demersal finfish permit were classified as set longlines (N = X), (3) vessels with longine and a tuna fishing permit were classified as pelagic longlines (N = X), (4) vessels with purse seines and a sardine/anchovy permit were classified as small pelagic purse seines, and (5) vessels with a purse seine and a tuna fishing permit were classified as tuna purse seiners.

We then determine whether a vessel is fishing based on speed thresholds defined in Table 1. Although imperfect, vessel speeds provide a low-cost solution to identify when a vessel might be engaged in fishing operations, but we note that this approach could also rely on predictions from a convolutional neural network [3], a gradient mixture model [8], or via a number of other computational approaches (CTE OTHER STUFF HERE). Because the clustering algorithm employed to identify fishing grounds can handle noisy data by excluding them from the cluster, false positives are not a large reason for concern.

2.3 Identifying Fishing Grounds

How DBSCAN works...

We define a fishing ground as a location with at least 5 hours of fishing activity on at least 5 distinct days per year. Since VMS pings are recorded at approximately hourly intervals, this translates to a base threshold of 25 pings per vessel-year of data. For each vessel, minPts $= 25 \times N_{\text{years}}$, where N_{years} is the number of distinct calendar years in the vessel's track. This scaling ensures that vessels with longer records must demonstrate sustained, repeated use of a location for it to qualify as a fishing ground.

We recognize that our 5-hours and 5-days is defensible yet arbitrary choice used to exemplify the application of the algorithm. In practice, the researcher would have the opportunity to derive this thresholds form previous work, or by directly interacting with the fishers. A

Table 1: Caption

Gear type	Speed range (knots)	N points	Source
Shrimp trawl	1 - 5		ADD CITATION HERE
Set longline	2 - 6		ADD CITATION HERE
Drifting longline	2 - 9		ADD CITATION HERE
Small pelagic purse seine	1 - 4		ADD CITATION HERE
Tuna purse seine	1 - 4		ADD CITATION HERE

robustness tests using different thresholds is presented in our Supplementary Materials section B.

The neighborhood radius (eps parameter) was then calibrated per gear type using per-vessel k-nearest-neighbour (k-NN) distance curves with $k = 25$. For each qualifying vessel (those with ≥ 50 pings), we computed the sorted k-NN distance curve and detected the elbow by identifying the maximum perpendicular distance from the line connecting the first and last points (Figure 1). The gear-level eps value is the median of all per-vessel elbow estimates. The resulting values are shown in Table 2.

Table 2: Neighborhood radius (eps, in km) by gear type. Statistics summarize the per-vessel elbow estimates from the k-NN distance curves. The median is the gear-level eps used to identify fishing grounds.

Gear type	Vessels (#)	Min	Median	Mean	Max	SD
Drifting longline	43	11.3	17.7	20.9	101.8	14.1
Set longline	170	4.5	12.4	15.3	123.7	13.6
Shrimp trawl	870	2.6	9.1	11.0	85.5	8.1
Purse seine	78	4.6	12.3	14.0	34.5	5.9
Tuna purse seine	51	25.7	141.0	141.3	215.3	39.2

2.4 Application 1: Space Utilization, Overlap, Re-visitation Rates, and Seasonality

2.5 Application 2: Exposure to Environmental Hazards

3 Results

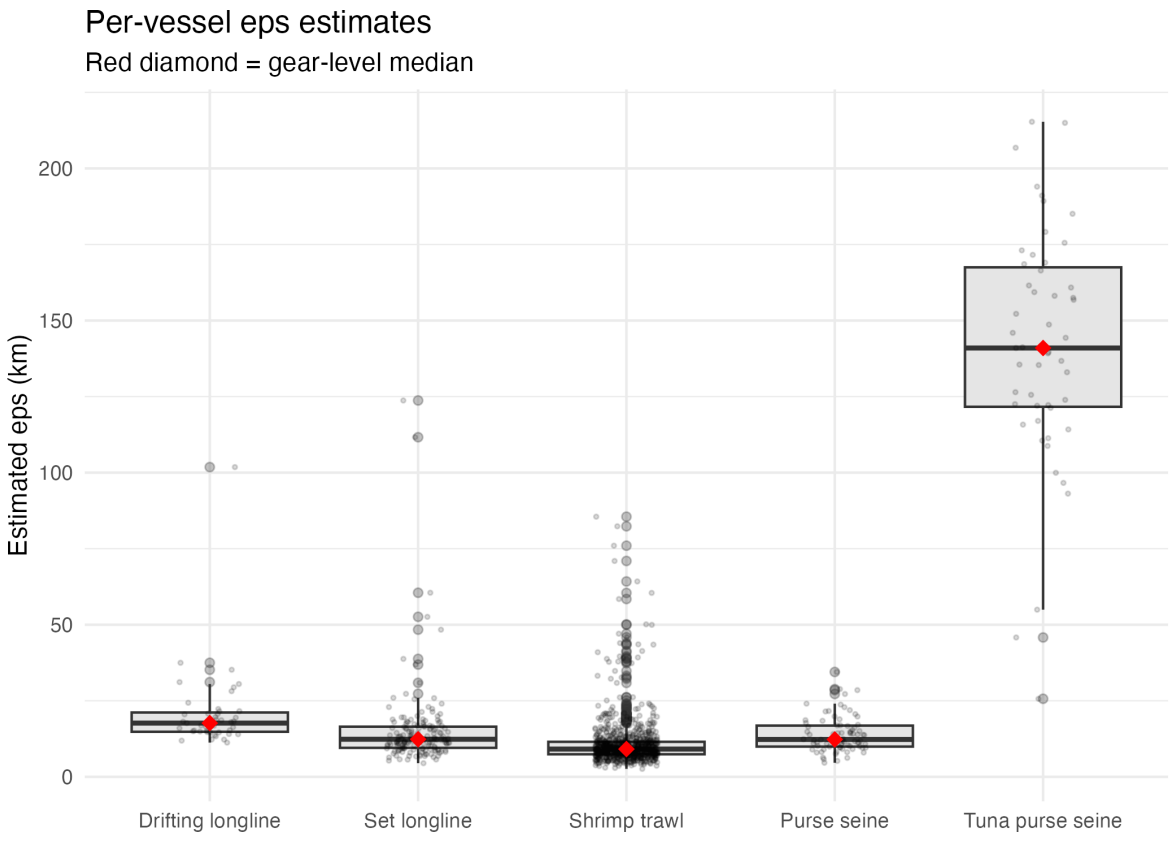


Figure 1: This figure will show how the knn neighbor is used to find the eps given a threshold of number of points.

Table 3: Summary of fishing grounds by gear type. Statistics are calculated across vessels with at least one fishing ground. Area and utilization are each vessel's totals across its fishing grounds. Points in grounds is the percentage of a vessel's VMS positions that fall inside its fishing grounds, as opposed to being classified as noise by DBSCAN. Gini coefficients measure how unequally fishing hours and area are distributed across fishing grounds.

Gear type	Vessels (#)	Total grounds (#)	Median grounds (#)	Median area (km^2)	Median utilization (hours)	Median points in grounds (%)	Gini utilization	Gini area
Drifting longline	40	148	4	22764.6	6746.7	30.5	0.67	0.60
Set longline	143	364	2	4980.3	3234.0	34.3	0.63	0.57
Shrimp trawl	862	3800	4	6402.3	11937.4	66.6	0.63	0.59
Purse seine	78	257	3	3524.4	4027.5	56.3	0.54	0.40
Tuna purse seine	49	82	1	164997.7	1413.3	19.5	0.47	0.46

3.1 Fishing Grounds

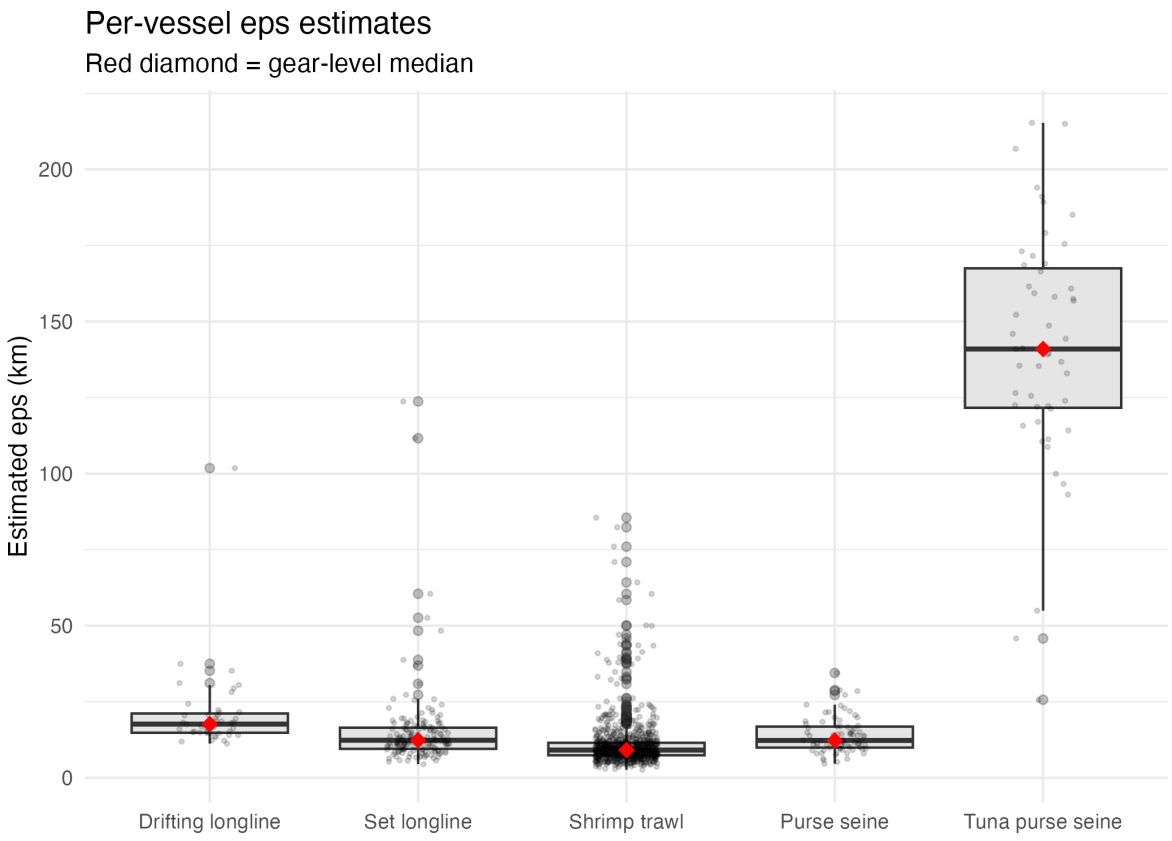


Figure 2: Caption

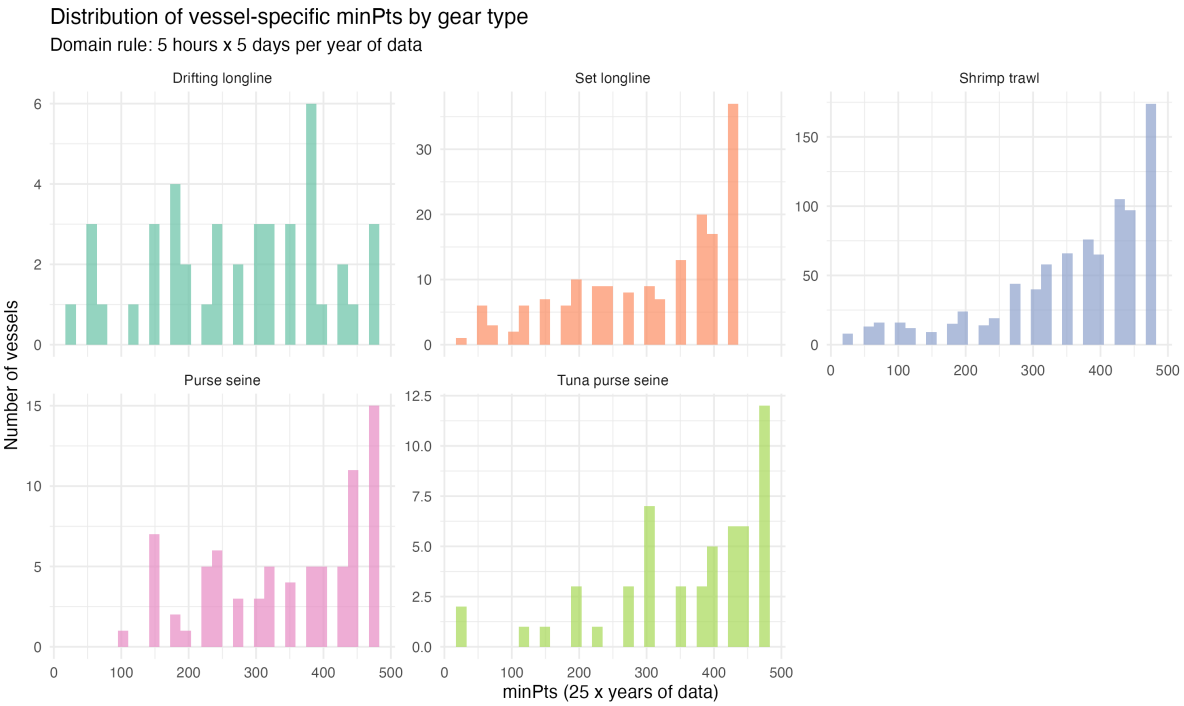


Figure 3: Caption

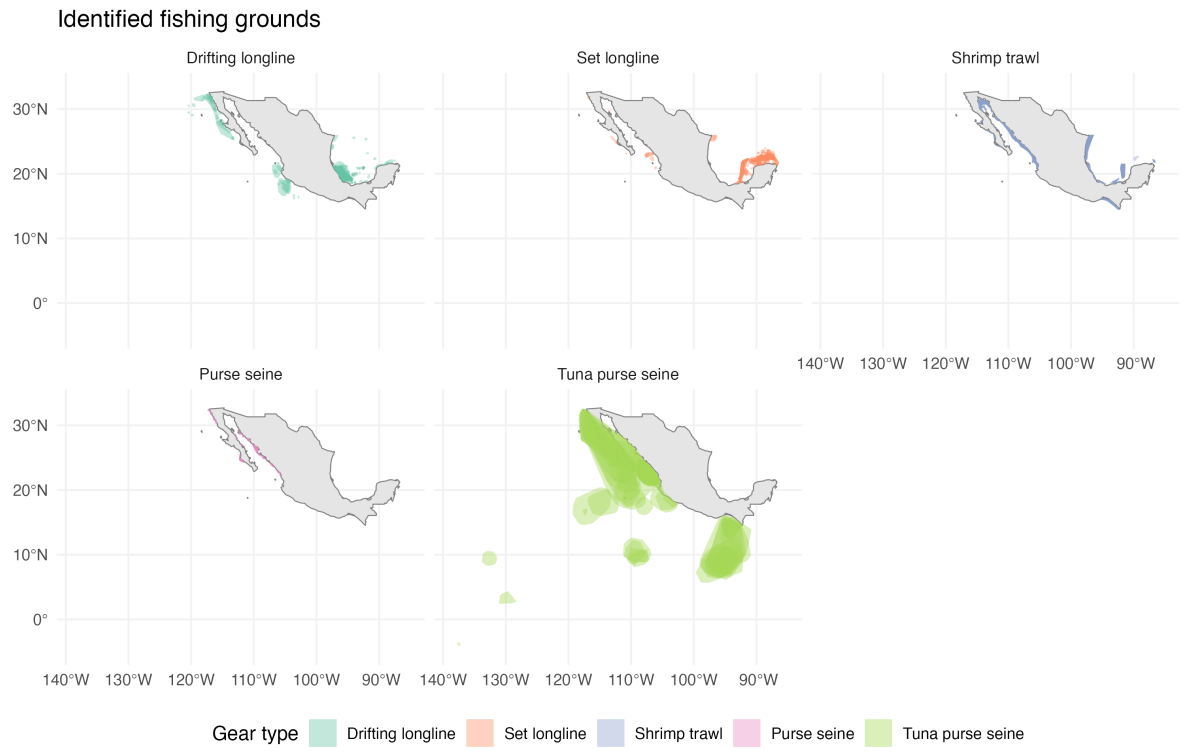


Figure 4: Caption

3.2 Application 1: Space Utilization, Overlap, Re-visitation Rates, and Seasonality

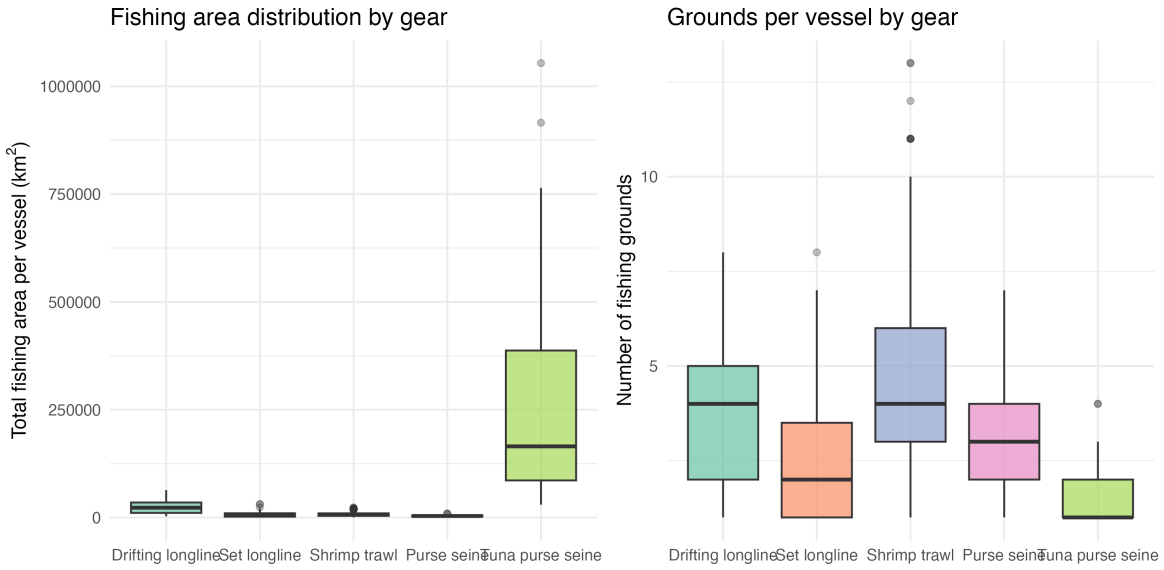


Figure 5: Caption

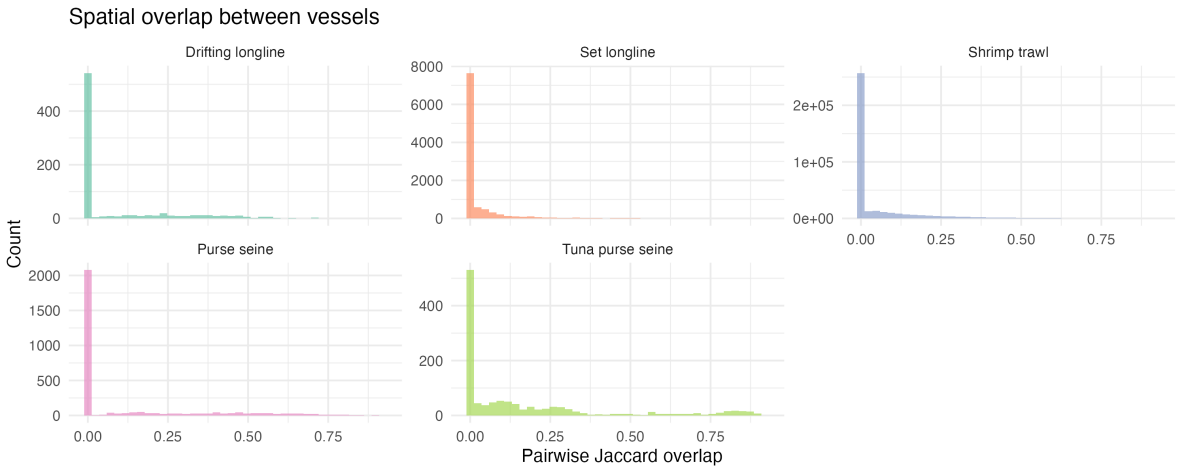


Figure 6: Caption

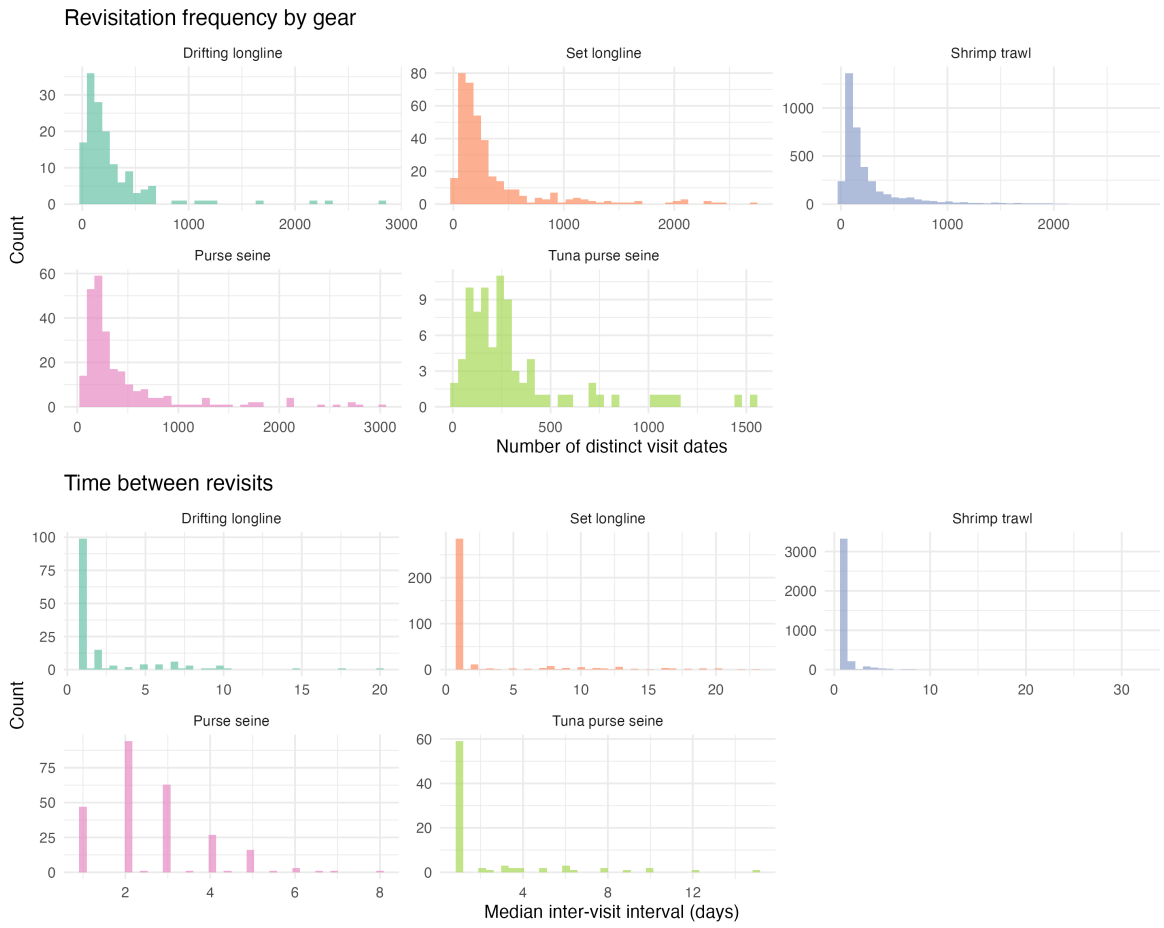


Figure 7: Caption

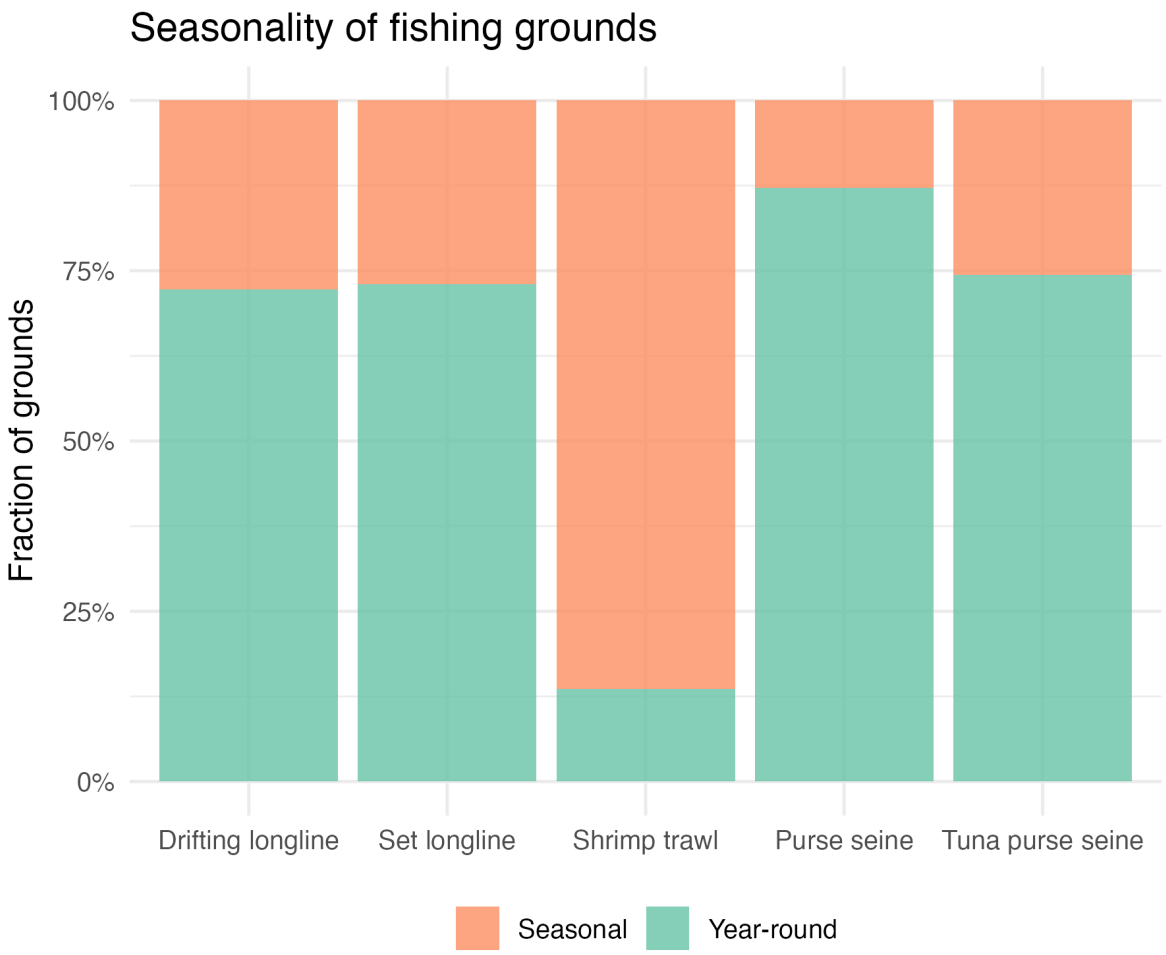


Figure 8: Caption

3.3 Application 2: Exposure to Environmental Hazards

4 Discussion and Conclusions

References

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Appendix

Supplementary Materials

For

Data-Driven Identification of Vessel-Level Fishing Grounds

A Other Figures and Tables

B Robustness to Other Minimum Utilization Thresholds